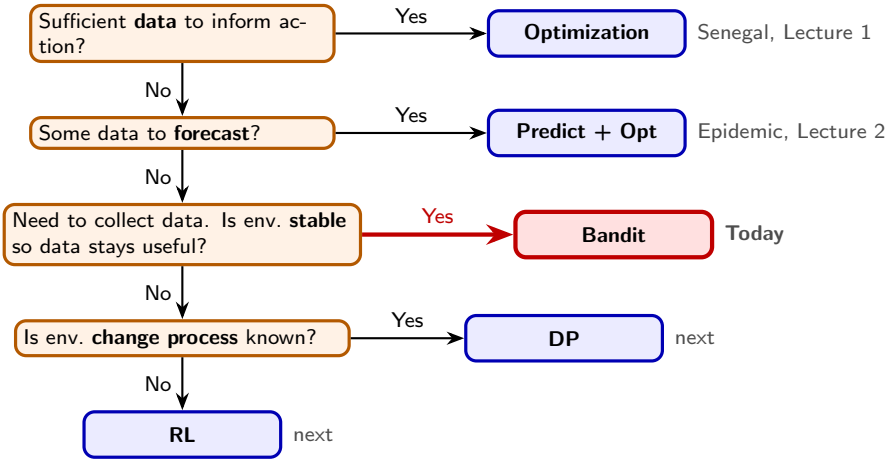
Lecture 3: Learning While Making Decisions

Multi-Armed Bandits, from Duolingo to a Bus in the Mon Valley

Peter Zhang

12768/19867/94867: From Data to Action

Where Today Sits



The new box: **the data does not exist, and the only way to get it is to act.**

Decisions that Serve Two Purposes

	Decision	What is unknown?	What one action reveals
Duolingo	Which reminder template should the next eligible learner receive?	The probability that each template brings a learner back to practice.	Show one template → observe whether that learner returns.
SNAP outreach	Should the next eligible household receive a text, phone call, letter, or navigator outreach?	The probability that each channel leads to a SNAP application or enrollment.	Use one channel → observe whether that household applies or enrolls.
Clinical treatment	Which accepted treatment should the next eligible patient receive?	The probability that each treatment achieves a predefined clinical outcome.	Give one treatment → observe the patient's subsequent outcome.

Every decision produces two things: (1) an outcome, and (2) information.

Example: Modern Products Keep Evolving After They Are Built

A traditional physical product is largely fixed when it leaves the factory.

Modern products are different: **the product keeps "improving".**

Product	Repeated decision	Business consequence
Google Search	Which ads to show, and where?	Advertising is a core revenue engine.
Facebook	Which content to show?	Engagement and advertising revenue.
Stitch Fix	Which items, messages, and product experiences should a client receive?	Conversion, retention.
Agentic systems	Which model, tool, or action should the system use next?	Quality, cost, engagement, task completion.

Designing the product → designing these repeated decisions.

Zoom In: Push Notification of an App

Consider one very common product interaction:

An app has an opportunity to send a user one notification per day.

The product team has several candidate messages:

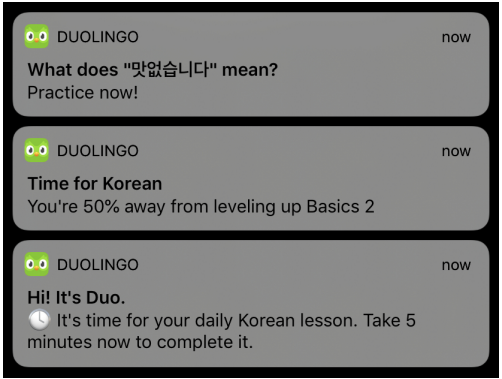
Option	Example
A	"Keep your streak alive."
B	"Five minutes is enough."
C	"Your friend just passed you."
D	"Your next lesson is ready."

For the next eligible user: the **system chooses one message** → the **user reacts**, for example returns or ignores it → the product records that outcome → another eligible user arrives, and the system chooses again.

Which message should the system choose next?

1. A product decision example
2. The Multi Armed Bandit Framework
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My Favorite Aggressive App



Yancey & Settles (2020), KDD.

- Duolingo picks one practice reminder per learner per day.
- Every reminder is **one template**, taken from a pool.

Assume you are the product manager designing this "picking" strategy.

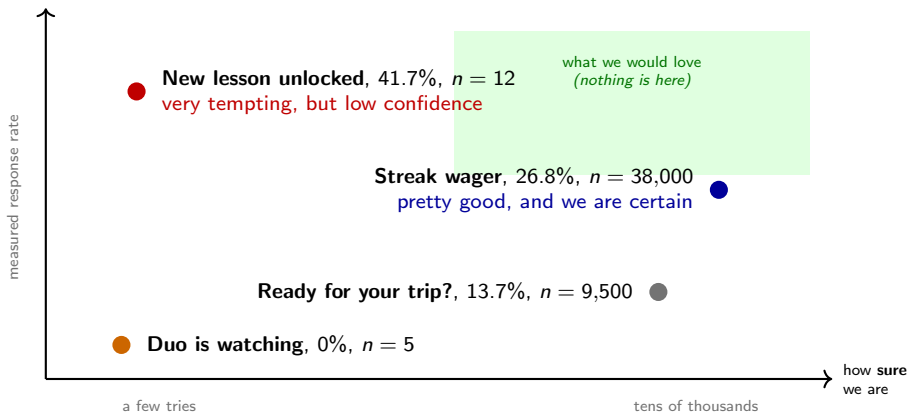
The Pool

Template	Text
You're on fire	Continue your 109 day Spanish streak.
Streak wager	Day 2 of your 7-day wager. Get to day 3.
Ready for your trip?	Take 5 minutes to practice Italian now.
Your friend passed you	Maria just overtook you this week.
5 minutes is enough	A short lesson still counts.
New lesson unlocked	Your next unit is ready.
Don't lose your XP	Your XP resets tonight.
Duo is watching	...

Data Collected So Far

Template	User clicked	Times picked by app
You're on fire	26.72%	40,000
Streak wager	26.82%	38,000
Ready for your trip?	13.72%	9,500
New lesson unlocked	41.7%	**12**
Duo is watching	0.0%	**5**

Two Different Things We Might Want



Simple Strategies Not Sufficient

Strategy 1

Always pick the template with the highest observed click rate.

Strategy 2

Pick each template equally often.

For each strategy:

What is good about it?

What might go wrong?

Some Plausible Strategies

1. **Mostly pick the current best, but occasionally try something else.**

   Example: 90% of the time pick the current winner; 10% of the time try another template.

2. **Prefer the current best, but deliberately give uncertain alternatives a chance.**

   Two natural ways to do this:

   - **Optimism:** give uncertain options an extra boost, because they could turn out to be better than they currently look.
   - **Randomized beliefs:** represent what values are still plausible for each option, then randomly choose according to those beliefs.

Same goal: get good outcomes now, but do not stop learning.

LIVE DEMO

Let's actually run it

`mab-bench.html` · tab 1, Duolingo

First: the translation table on the left. That is the entire vocabulary of this lecture, attached to a problem you have already argued about.

Then: run *always pick the current best*, press *Keep curve*, run *score by best-it-could-be*. Watch the red whiskers: tall on the templates picked only a few times, short on the ones picked often.

Finally: tick *reveal the truth*. Then press *New draw* and see whether the same thing happens twice.

What Is That Difference Worth?

Measured in the bench, over 3,000 pushes, averaged over 24 runs and 40 worlds:

Rule	Click rate	vs. ϵ-greedy
Rotate at random	0.2365	
Always the best so far	0.2803	
ϵ-greedy, $\epsilon = 0.1$	0.2837	
UCB, $c = 0.2$	**0.2912**	**+0.0075** (+2.6%)

30 million learners, one reminder each per day	10.9 billion picks a year
$\times$ 0.0075 extra clicks per pick	**82 million** extra practice sessions
$\times$ about \$0.06 per practice session	$\approx$ **\$4.9 million a year**

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Exploitation and Exploration

Exploitation

Stick with what is known.

Safely collect some reward.

Exploration

Keep trying the uncertain ones.

Maybe find something better.

Every action pays you **now**

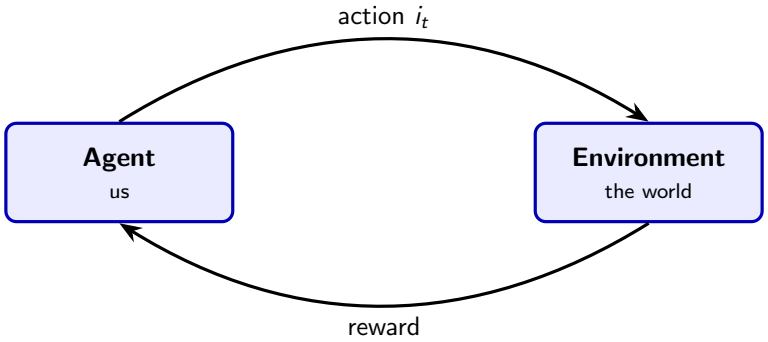
and teaches you something for **later**.

What was exploited and explored in the previous example?

And You Solve This Every Week

Setting	Exploitation	Exploration
Job hunting	Take the current offer	Sit through more interviews
Lunch	The place you always go	The new place on Craig St
A product team	Improve the existing product	Build a new one
Political campaign	Run the ad that works	Test a new message
Medical treatment	The well-tested drug	The trial drug

General Setup



The environment does not immediately hand us a dataset.
We will need "learning while doing".

In Lectures 1 and 2, data was already obtained from some prior work or prediction.

Where the Name Comes From

[not covered in class]



- A **one-armed bandit** is slang for a slot machine: it robs you slowly.
- A row of them, each paying at a different unknown rate, is a **multi-armed bandit**.
- Fixed number of pulls. Which machine, how many times?

Multi-Armed Bandit

Each arm a pays 1 with some true probability p_a, and 0 otherwise.

If every p_a were known:

$$i_t = \arg\max_a \; p_a \qquad \text{for every } t. \qquad \text{Done.}$$

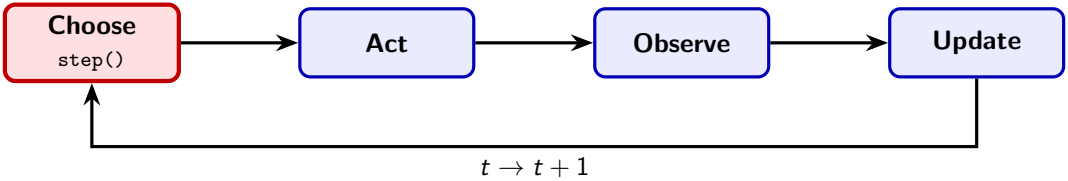
But p_a is not known. All we ever have is

r_a^t = the average reward we have seen from a, $\qquad$ n_a^t = how many observations that is.

State Information

We carry and update these two numbers over time.

r_a^t = average reward from a so far, $\quad n_a^t$ = times we picked a



The strategies change "Choose".

Three Rules for `step()`

Rule	**What it picks**
Random	Any arm, uniformly. Uses nothing it has learned. The baseline.
ϵ-Greedy	$\arg\max_a r_a^t$ with probability $1-\epsilon$; a random arm with probability ϵ.
UCB	$\arg\max_a \left[r_a^t + c\sqrt{\frac{\ln t}{n_a^t}} \right]$, the estimate *plus* a margin for what we do not know.

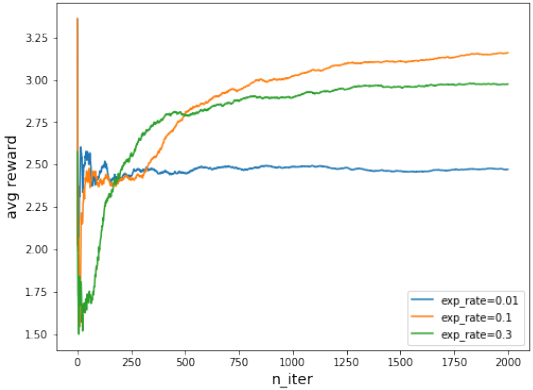
What Does ϵ Buy You?

ϵ = the fraction of traffic you are willing to spend on learning.

Questions:

- $\epsilon = 0$. What goes wrong?
- After a year at $\epsilon = 0.3$ you *have* found the best arm. Are you done?
- Should ϵ be the same on day 1 and on day 500?

Simulated



- $\epsilon = 0.01$ flattens early and low. It committed.
- $\epsilon = 0.3$ finds the best arm, then keeps paying a 30% tax forever.
- $\epsilon = 0.1$ wins here. Nobody could have told us that in advance, which is itself an argument against ϵ-greedy.

We Are Throwing Away n_a^t

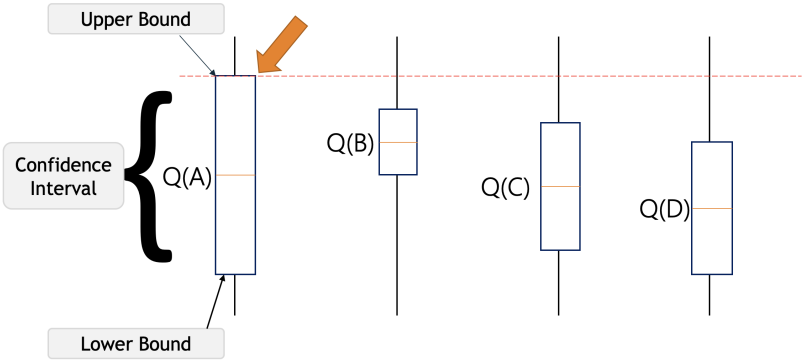
- So far only r_a^t has been used. But we also know **how many observations each average rests on**.
- More trials means more confidence, which means a **smaller margin of error** around the average.

Optimism in the face of uncertainty.

Score every arm by *the best it could plausibly be*. Pick the winner.

An arm gets picked if it **looks good**, or if we are **unsure**. Both are good reasons.

Picture It



UCB picks arm A, not because its average is best, but because its *upper bound* is.

That is "New lesson unlocked, 41.7% on 12 tries", drawn properly.

Upper Confidence Bound

$$i_t = \arg\max_a \Bigg[\underbrace{r_a^t}_{\text{what we believe}} + \underbrace{c\sqrt{\frac{\ln t}{n_a^t}}}_{\text{what we do not know}} \Bigg]$$

- n_a^t large means a small margin. We already know this arm.
- t grows while n_a^t does not, so the margin **grows**. An ignored arm becomes attractive again *just because time passed*.
- $n_a^t = 0$ makes the margin infinite. **Every arm gets tried at least once, automatically.**

→ **A1 Written 3.1(b) and 3.1(c) are the second and third bullets.**

How Do You Actually Set c?

1. Start from theory, then leave it. UCB1 proves its bound at $c = \sqrt{2}$ for rewards in $[0, 1]$. That is a *worst case* guarantee, not a recommendation, and it over-explores at short horizons.

2. Rescale to your reward. The bound assumes $[0, 1]$. For rewards in $[0, R]$, use $c \cdot R$. Dollars and clicks do not take the same c.

3. Sweep it offline, on logged data. Replay your own logs and score each c before any user is touched. This is what Duolingo did for its τ. Then sweep again in a simulator. **Take the knee, not the peak**; the peak is fitted to your simulator.

4. Check the explore rate it implies. "What share of traffic goes to something other than the current best?" is the number a stakeholder can approve.

5. Re-tune when the horizon changes. The best c depends on T.

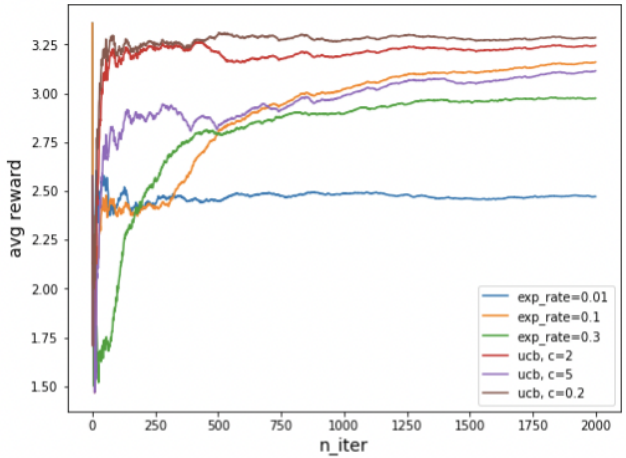
Sweep from the bench, $T = 3{,}000$:

c	click rate
0.05	0.2829
0.10	0.2828
0.20	**0.2905**
0.30	**0.2908**
0.50	0.2803
0.80	0.2692
1.41 ($\sqrt{2}$)	0.2572
2.00	0.2519

The textbook value is the **worst** row on this table.

And if you find yourself tuning c a great deal, that is an argument for **Thompson sampling**, which has no c.

All Three, Side by Side



- UCB climbs fast **and** keeps climbing.
- Its exploration is **targeted**. It goes to the arms we know least about, not to a random one.
- **UCB is not the only good answer. Thompson sampling**, draw a plausible rate for each arm and pick the winner of that draw, has **no knob at all** and catches up as the horizon grows. It is in the demo; try it.

→ **A1 3.2(e): this relates to the figure you hand in.**

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Be the Algorithm

Four arms, each already picked once, so $t = 4$ is done:

Arm a	1	2	3	4
r_a^t	0.95	0.88	0.60	0.70
n_a^t	1	1	1	1

The rewards waiting at each step. **You only ever see the one you pick:**

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	0.85	0.92	0.80	0.75
$t = 6$	0.72	1.00	0.68	0.70
$t = 7$	0.80	0.90	0.65	0.55
$t = 8$	0.75	0.85	0.71	0.60
$t = 9$	0.70	0.98	0.64	0.67

Take $c = 0.1$. **Which arm is i_t for $t = 5$ through 9?**

Step $t = 5$

$\ln 5 = 1.609$, every $n_a^t = 1$, so every "upper confidence" margin is $0.1\sqrt{1.609} = 0.127$

Arm	r_a^t	n_a^t	margin	**score**
1	0.950	1	0.127	**1.077**
2	0.880	1	0.127	1.007
3	0.600	1	0.127	0.727
4	0.600	1	0.127	0.727

the reward matrix

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	**0.85**	0.92	0.80	0.75
$t = 6$	0.72	1.00	0.68	0.70
$t = 7$	0.80	0.90	0.65	0.55
$t = 8$	0.75	0.85	0.71	0.60
$t = 9$	0.70	0.98	0.64	0.67

$i_5 = 1$. We see 0.85, and nothing else. $n_1 = 2$, $r_1 = \frac{0.95+0.85}{2} = 0.900$.

When every arm is equally well known the margin cancels, and UCB is simply "pick the best so far".

Step $t = 6$

$\ln 6 = 1.792$. Arm 1's margin just shrank.

Arm	r_a^t	n_a^t	margin	**score**
1	0.900	2	0.095	0.995
2	0.880	1	0.134	**1.014**
3	0.600	1	0.134	0.734
4	0.700	1	0.134	0.834

the reward matrix

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	**0.85**	0.92	0.80	0.75
$t = 6$	0.72	**1.00**	0.68	0.70
$t = 7$	0.80	0.90	0.65	0.55
$t = 8$	0.75	0.85	0.71	0.60
$t = 9$	0.70	0.98	0.64	0.67

$i_6 = 2$. We see 1.00. $\quad n_2 = 2, \quad r_2 = 0.940.$

Arm 1 has the *higher average* and loses anyway. Being better measured is a cost.

Step $t = 7$

$\ln 7 = 1.946$

Arm	r_a^t	n_a^t	margin	**score**
1	0.900	2	0.099	0.999
2	0.940	2	0.099	**1.039**
3	0.600	1	0.139	0.739
4	0.700	1	0.139	0.839

the reward matrix

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	**0.85**	0.92	0.80	0.75
$t = 6$	0.72	**1.00**	0.68	0.70
$t = 7$	0.80	**0.90**	0.65	0.55
$t = 8$	0.75	0.85	0.71	0.60
$t = 9$	0.70	0.98	0.64	0.67

$i_7 = 2$. We see 0.90. $\quad n_2 = 3, \quad r_2 = 0.927.$

Arm 2 is now the best on both counts, so it keeps winning. Its margin is shrinking as it does.

Step $t = 8$

$\ln 8 = 2.079$

Arm	r_a^t	n_a^t	margin	**score**
1	0.900	2	0.102	1.002
2	0.927	3	0.083	**1.010**
3	0.600	1	0.144	0.744
4	0.700	1	0.144	0.844

the reward matrix

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	**0.85**	0.92	0.80	0.75
$t = 6$	0.72	**1.00**	0.68	0.70
$t = 7$	0.80	**0.90**	0.65	0.55
$t = 8$	0.75	**0.85**	0.71	0.60
$t = 9$	0.70	0.98	0.64	0.67

$i_8 = 2$. We see 0.85. $\quad n_2 = 4, \quad r_2 = 0.908.$

Arm 2 wins by **0.008**. Its average is falling, its margin is shrinking, and arm 1 is catching up without being touched.

Step $t = 9$

$\ln 9 = 2.197$

Arm	r_a^t	n_a^t	margin	**score**
1	0.900	2	0.105	**1.005**
2	0.908	4	0.074	0.982
3	0.600	1	0.148	0.748
4	0.700	1	0.148	0.848

the reward matrix

Step	Arm 1	Arm 2	Arm 3	Arm 4
$t = 5$	**0.85**	0.92	0.80	0.75
$t = 6$	0.72	**1.00**	0.68	0.70
$t = 7$	0.80	**0.90**	0.65	0.55
$t = 8$	0.75	**0.85**	0.71	0.60
$t = 9$	**0.70**	0.98	0.64	0.67

$i_9 = 1$. We see 0.70, a bad draw. $n_1 = 3$, $r_1 = 0.833$.

Arm 2 had the higher average, and we just lost by switching. Did UCB make a mistake?

What the Five Steps Showed

$$i_5 \ldots i_9 \;=\; 1 \to 2 \to 2 \to 2 \to 1$$

- It found arm 2 and stayed with it. *exploitation*
- It went back to arm 1 because that margin had grown. *exploration, with no coin flip*
- The bad draw is the price. It is also what buys the information.

Arms 3 and 4 were never touched again. Feature or bug?

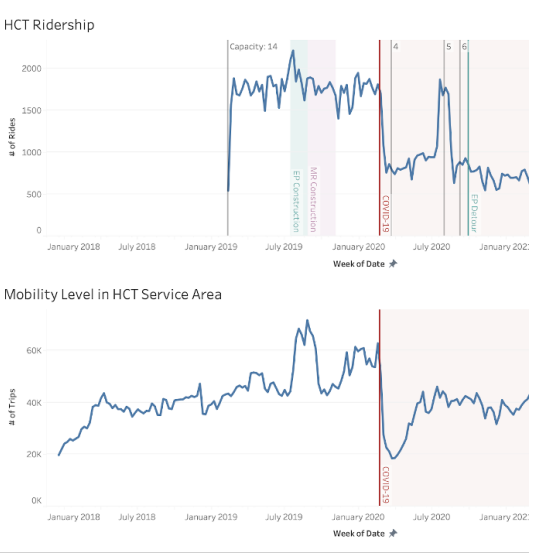
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Heritage Community Transportation



- Nonprofit transit in the **Mon Valley**, Allegheny County, 40 minutes from this room.
- Rides to jobs, groceries, clinics.
- Fares: **25 cents**, and 10 cents for children and riders with disabilities.
- Funded by state subsidies, county contracts and philanthropy, all of which want to see **ridership**.

Their Problem, in One Chart



- Ridership **halved** after March 2020 and never returned.
- Mobility in the same area **did** return. The riders are moving, just not on the van.
- The number their board sees:

$$\text{cost per rider} = \frac{\text{total cost}}{\text{number of riders}}$$

Exercise: You Are Advising HCT

Outreach options, one small budget:

flyers at churches · SMS to the rider list · local Facebook groups · radio spots · employer partnerships

Fill in five rows.

Step t	?
Arm a	?
Reward	?
Cost c_a	?
Objective	?

Different Ways to Define Reward

a signup? a first ride? rides per month?

still riding in six months? minus cost per rider?

- Reward = signups, and it optimizes for people signing a form. Some never ride.
- Reward = rides, and it chases heavy users and abandons first-timers.
- Reward = retained riders. Right, and unobservable for six months.

Choosing the reward is the modeling. It is Lecture 1's "min ?" in different clothes.

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The Abstraction Is Deliberately Thin

To make one algorithm work everywhere, we quietly assumed:

1. Every arm is **always available**.
2. Each arm's value is **fixed**. Picking it does not change it.
3. Every pick **costs the same**.
4. The reward arrives **immediately**, and the arm is all that matters, **not who is in front of you**.

Duolingo Breaks 1 and 2; HCT Breaks 3

Sleeping

"109 day streak" needs a 109 day streak.

Arms are eligible only for some learners, who click more at *everything*.

A restricted arm looks great for reasons unrelated to the message.

Recovering

A template works better *because* it is new.

It decays as you use it and heals while it rests.

The harder you exploit a winner, the more you destroy what you measured.

Cost

A \$12 employer partnership and a \$0.05 SMS are the same "pick".

One division fixes it:

$$\arg\max_a \frac{r_a^t + c\sqrt{\ln t / n_a^t}}{c_a}$$

And you stop when the money runs out, not the steps.

Sleeping and recovering are a **state**, a tiny one, but a state. **That is the door into next week.**

Extended MAB You Will See in Practice

Name	The tell, how you recognise it	What changes
Contextual	"It depends who it is." The best arm differs between people.	Keep r_a^t *per segment*, or fit a model $r_a(x)$.
Non-stationary	"That used to work." The world drifts, or a shock arrives.	Replace the plain average with a discounted or sliding-window one.
Delayed feedback	"We won't know for months." The outcome lands long after the action.	Update late, or act on a fast proxy and reconcile later.
Budget / knapsack	"These cost wildly different amounts." A pick is not a pick.	Divide the score by c_a; the horizon becomes a budget.
Batched	"We ship on Mondays." You cannot re-decide per person.	Choose a *mix* over arms; update once per batch.
Sleeping / recovering	"Not everyone is eligible"; "it wears out."	Mask ineligible arms, score relatively, subtract a recency penalty.
Constrained / fair	"We are not allowed to starve that group."	Add a floor per arm or group, or raise the effective cost of over-served ones.

Every row is a change to `step()` or to the definition of the reward. The loop does not change.

Exercise: Match the Situation to the Structure

Situations

A. A hospital picks among 4 discharge-reminder scripts. The best script differs substantially by age, language, and diagnosis.

B. A city tests 6 pothole-reporting prompts. After a major news story, residents become much more likely to report potholes regardless of the prompt.

C. A clinic can contact patients by SMS, phone call, or home visit, costing \$0.20, \$4, and \$30, under a fixed annual outreach budget.

D. A scholarship program tests several recruitment channels, but whether a contacted student enrolls is not known until months later.

E. A job-training nonprofit optimizes outreach but must ensure at least 15% of contacts go to rural applicants.

Structures

1. Delayed-feedback bandit
2. Contextual bandit
3. Constrained / fair bandit
4. Non-stationary bandit
5. Bandits with knapsacks

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The Checklist

When you meet a new problem, in this order:

1. What happens in one **step**?
2. What are the **arms**?
3. What is the **reward**, and when do you see it?
4. What does a pick **cost**?
5. Which of the **four assumptions** does this problem break?

Random $\rightarrow$ ϵ-Greedy $\rightarrow$ UCB / Thompson $\rightarrow$ the extension your problem needs

In that order. Duolingo went all the way right, *after* showing the first three did not work.

The Most Important Choice Is Often the Reward

For HCT, suppose the algorithm is choosing outreach activities.

What should count as "success"?

Reward	**What the algorithm will learn to pursue**
Signup	People who fill out a form
First ride	Getting new riders onto the service
Number of rides	Frequent riders and high-volume usage
Still riding after 6 months	Long-term adoption
Rides per dollar	Cost-efficient ridership growth

The algorithm learns what you ask it to learn.

Choosing the reward is therefore a substantive modeling decision, not a technical detail.

Taking This to Your Project

Across three lectures, the hard part was not the method:

	Where the difficulty actually was	**So, for your project**
Senegal	Not the solver. The objective: whose distance counts, and who gets left behind.	Write your objective in one sentence. Then write a different one that is also defensible. If you cannot, you do not have a project yet.
Epidemic	Not the curve fit. The mechanism, and getting it in front of the Governor's office by Wednesday.	Scope to what you can finish. A rough answer on time beats a good one afterwards.
Bandits	Not the algorithm. What one step is, what the arms are, and what the reward should be.	Name the decision maker, the action set and the reward before you write any code.

- **Can you name who acts differently because of your work?**
- **Can you actually get the data?** Senegal spent three years on measurement. If you cannot get it, simulate it and be transparent.
- **Show a trade-off, not a number.** Cohon's point: the model lays out the policy space, but it does not end the argument.

Taking This to the Job Search

	What the role is called	What the posting says
Senegal	Operations research, decision science, strategy analytics	optimization, integer programming, facility location, resource allocation
Epidemic	Data science, forecasting, policy modeling	simulation, calibration, scenario analysis, capacity planning
Bandits	Experimentation, personalization, growth, applied ML	A/B testing, adaptive experimentation, off-policy evaluation, recommender systems

- **Build one small end-to-end thing you can demo.**
- **Lead with the decision problem, not the method.**
- **Be ready for "what if that assumption is wrong?"**
- **The scarcity: translate between the domain and the methodology.**